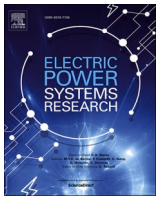
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Synchronization and control of WECS-SPV-BSS-based distributed generation system using ICCF-PLL control approach

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ABSTRACT

The present paper suggests a synchronization and control strategy of a distributed generation (DG) system. An improved phase-locked loop (ICCF-PLL) control approach based on a complex coefficient filter is also described. An ICCF-PLL control approach provides a fast dynamic response, reduces steady-state error, and minimizes oscillations under recurring conditions of renewable energy sources (RESs) and load perturbation. A Voltage Source Converter (VSC) using the ICCF-PLL control approach grants voltage/frequency stabilization, power support, load balancing, mitigation of harmonics introduced by non-linear loads and helps in improving the power quality (PQ) in the DG system. In the proposed DG system, a solar photovoltaic (SPV) array is interfaced to the DC-link, and a modified Perturb & Observe (MP&O)-based tracking method is being utilized to draw maximum power from an SPV. Further, a standalone DG system suffers from poor voltage and frequency regulations during load perturbation; therefore, for maintaining the power equilibrium between the power generator and loads, a battery storage system (BSS) is linked on the DC-link side of the VSC through a bidirectional DC-DC converter (BDC). Complete mathematical modeling of the ICCF-PLL control approach is demonstrated and a Simulation/Experimental study is performed to evaluate the proposed system operation.

1. Introduction

Renewable energy sources (RESs) are alternative options to fulfill the increasing need for energy demands of both rural and urban populations. Renewable energy development and utilization should be prioritized, because of increasing concern for environmental pollution due to coal and petroleum-based generation. The worldwide sustainable energy development is expanding significantly by stronger support from government policies and addressing key barriers such as permitting higher level grid integration of RESs. The growth of RESs (wind and solar) in both developing and developed countries is critical for achieving energy sector sustainability [1]. The utilization of RESs for energy production had become a favored alternative due to substantial cost reduction over the past decade's advancement in technology and reduction in home gas emissions. Due to vast availability, environmentally benign nature, and reduction in the cost of solar PV panels, solar-based power production had seen fantastic growth in contrast to other power generation sources during the last decade. The main disadvantage of solar energy was that it had intermittent nature. As a result, the SPV cannot meet the load requirement at all times [2,3]. This issue was solved significantly by combining a battery system [4,5] or ultra-capacitor with a PV array [6]. A Variable Kernel Width Maximum Correntropy Criterion (VKW-MCC)-based approach had been utilized to manage the uncertainty condition of RESs and load [7]. An energy storage system had been utilized to provide power support and voltage stability under different operating conditions [8] and a sample average approximation approach had been also utilized to operate the hybrid microgrid [9]. A fuzzy number had been used to reduce the uncertainty condition of RESs [10]. A Multi-Agent System (MAS) theory was described to provide robust control in an islanded microgrid [11]. A non-linear adaptive second-order Volterra filter (NVLF)-based control with VSC was implemented to enhance the overall PQ of distributed generation (DG) systems [12].

An improved Gaussian filter-based PLL [13] procedure had been established to overcome the PQ problems and provides harmonic compensation, voltage/frequency regulation, and power support during the recurring situations of RESs and load perturbation [14]. An improved P&O control approach [15] was described to draw the maximum power from an SPV during the unreliable situation of solar intensity [16,17]. A model predictive control technique was designed to

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generate pulses for a 3-phase inverter and this control evaluates the predictions for upcoming values [18]. Moreover, a Quasi–Newton Least Mean Fourth (QNLMF)-based control had been designed to generate reference current for pulses generation of VSC and provides voltage stability in wind-based off-grid 3-phase 4-wire DG systems [19]. A Modified Second Order Sequence Filter (MSOSF)-based FLL control had been utilized to boost synchronization and continuous mode switching response by predicting the frequency and phase angle rapidly and correctly under unbalanced and distorted grid currents [20,21]. An LMS-based control technique for DSTATCOM with the DC-offset rejection ability to provide a precise transient response with minimal error was described [22]. In dynamic situations, an in-loop filter-based ESOGI-FLL control approach had been developed to monitor the frequency and tune to frequency swings [23]. A microgrid had been designed for both islanded and grid interface modes during a reliable synchronization control. The microgrid allows charging for Electric Vehicles (EVs), and it balances and manages the common point of inter-section (CPI) voltage profile and frequency in an islanded mode with residential and dynamic loads. In an isolated mode of microgrid, the CPI frequency and voltage are often adjusted under the conditions of a rapid commencement of an IM load, unsteady non-linear loads, and EV charging [24]. Transmission delays and network transmission methods had also been employed, as is a distributed model-based dynamic event-triggered transmission methodology [25]. The use of Artificial Intelligence technology appears to be a potential alternative for improving microgrid operation and control in forthcoming smart grid networks [26]. Coordinated control and reliability of distributed energy sources [27] and the need for reliability-oriented design were especially important for microgrids that utilize a significant share of non-conventional energy-based distributed energy resources (DERs) [28]. An all-pass filter-based PLL was described to pass all components of frequency without transforming their peak value but shifting their phase 90° and Complex Coefficient Filter (CCF)- based PLL was also utilized with harmonic rejection capability [29]. While maintaining appropriate dynamic behavior, the CCF-PLL rejects/extracts disruptions before the PLL control loop [30]. A WECS-SPV-BSS-based DG system is becoming very prominent as a standalone power system for generating power in islanded locations and hilly terrains, deserts, and remote places where grid supply is not available [31,32].

The paper is arranged in the following manner: The System Description is briefly laid out in Section 2. Its Schematic diagram is shown with all of the details. The Control Methodology of the Isolated DG System is addressed in Section 3. Section 4 presents the Results and Discussion of the dynamic response of the proposed isolated DG system. Section 5 presents the Test Results, and Section 6 presents the research paper's outcomes and findings.

In this paper, a three-phase DG system based on WECS, SPV, and BSS feeding unbalanced nonlinear loads is intended to provide the following improvements in DG systems.

a) An ICCF-PLL control approach is significantly reduced the ripple magnitude in fundamental load current and hence improves harmonics eradication capability.
b) An ICCF-PLL is successfully able to evaluate the magnitude of the fundamental component of load current which provides improved convergence with reduced oscillations and better noise eradication with the reduced oscillations in comparison to existing controls.
c) The proposed control with VSC contributes active/reactive power support, voltage/frequency regulation at PCI, abatement of harmonics, load balancing, and reduction in overall PQ issues of the DG system under changing conditions of RESs and unbalanced load.
d) An ICCF-PLL control approach successfully detects the frequency of the DG system and acclimates to the frequency fluctuations during intermittent situations. The total harmonic distortions (THDs) in the source (voltages & currents) of the proposed DG at PCI are consistent with the IEEE-519 standard.

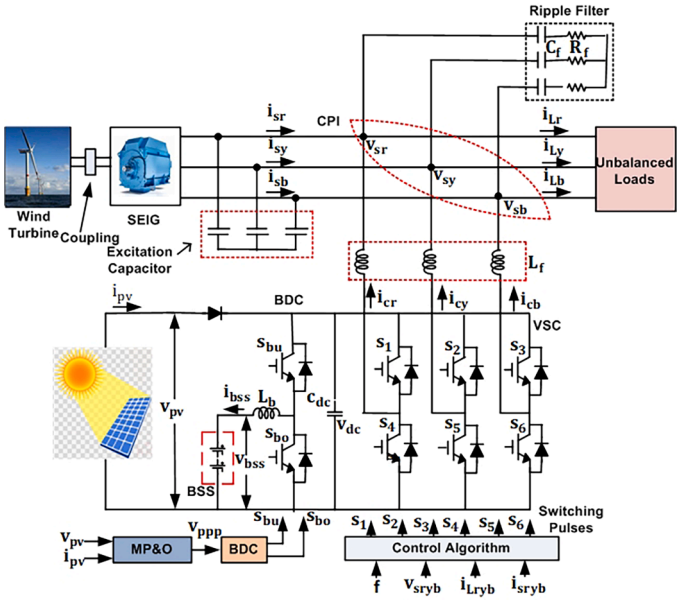


Fig. 1. Schematic diagram of DG system.

e) An MP&O control algorithm is employed to extract maximum power from an SPV during intermittency of solar insolation. Furthermore, a BDC controller is intended to manage the charging and discharging modes of BSS. Under different operating conditions, the BSS maintains the power balance between the power generation sources and the load.

2. System description

The implemented design of the proposed DG system is depicted in Fig. 1. The DG system consists of a Wind Energy Conversion System (WECS) of 3.7 kW, SPV of 2.5 kW, a battery storage system (400 V, 7.5 Ah), interfacing inductors, and nonlinear load. An MP&O-based Maximum Power Point technique (MPPT) is utilized to extract maximum power from a PV array under the random behaviors of solar irradiations. Further, a BDC control-based battery system is utilized to store unused power during the abnormal condition of RESs and wobbly loading. Moreover, the BSS supplied the required power at PCI with the help of BDC control and establish the power equilibrium among the RESs and the load. Interfacing inductors are utilized to reduce the current ripples caused by VSC. A parallel connected ripple filter is also utilized to attenuate high-frequency switching noise [15].

3. Control methodology of isolated DG system

The control techniques in a standalone system offer voltage/frequency stabilization, and mitigation of harmonics, along with a balance between active and reactive power under load perturbation. The proposed control strategy consists of SPV-based MPPT control, Bi-directional control of the DC-DC converter for charging and discharging of BSS, and a VSC control scheme based on ICCF-PLL control. Implementation of ICCF-PLL control provides a fast dynamic return with minimized oscillations in DG system operation and extract fundamental load current component during varying conditions of RESs and load. The control structure of the standalone DG system is depicted in Fig. 2. An MP&O control algorithm is designed and implemented to harness maximum power from the SPV system during differing conditions of solar energy [17].

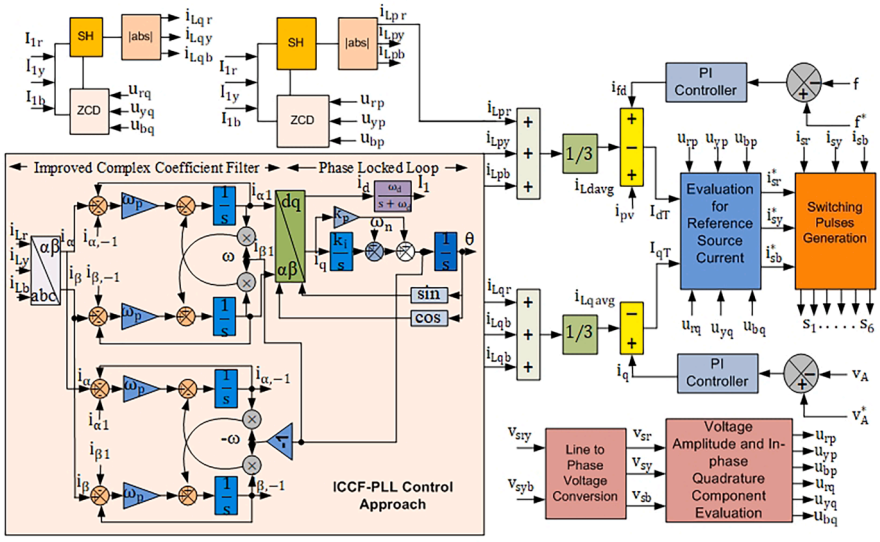


Fig. 2. Control scheme of DG system.

3.1. MP&O-based MPPT control

The proposed MP&O technique extracts maximum power point (MPP) voltage from an SPV during a variation in solar insolation. The duty ratio of the converter is estimated using the SPV output voltage from the MP&O-based MPPT approach and the observed DC-link voltage. The following steps are taken for the MP&O-based MPPT approach.

$$\left(\text{if } \Delta P_{pv} > 0\right)\text{then } \Delta P_{pv}(s) = P_{pv}(s) - P_{pv}(s-1) \tag{1}$$

$$\left(\text{if } \Delta P_{pv}(s) > \Delta P_{pv}\right) \text{ then } v_{s\ new}(s-1) = v_{sold} + \Delta v_{pv} \tag{2}$$

$$\left(\text{if } \Delta P_{pv} < \Delta P_{pvold}\right) \text{ then } v_{s\ new} = v_{s\ old} - \Delta v_{pv} \tag{3}$$

Where, (ΔP_{pv}) and (Δv_{pv}) are represent the variation in SPV power and SPV voltage. $(v_{s\ new})$ is the maximum power of the PV array and (Δv_{pv}) is the step size for the PV voltage (v_{pv}) [15].

3.2. Bi-directional control

In a solar-wind hybrid DG system situation of real power unbalance arises due to changes in wind speed/solar intensity and load perturbation, which disturbs the control of DC-link voltage. Further, intermittency in RESs disturbs the PQ such as the frequency & voltage of the DG system.

Thus, regulation of DC-link voltage performs an important role in the DG system and it facilitates stabilization of the frequency and voltage at PCI. In this paper, a BDC controller is used for the regulation of the voltage at the DC-link, which consists of two PI controllers.

The controller-I is written as for the regulation of DC-link voltage.

$$i_{bss}(s) = i_{bss}(s-1) + k_{bp}\{v_{dce}(s) - v_{dce}(s-1)\} + k_{bi}v_{dce}(s) \tag{4}$$

Where DC-link voltage error is evaluated as

$$v_{dce}(s) = v_{mpp}(s-1) - v_{dc}(s) \tag{5}$$

Where, (v_{mpp}) is defined as the reference voltage of the DC-link computed through an MP&O-based MPPT algorithm.

The controller-II output is utilized to compute the operation cycle (ρ) of the BDC, it is described as

$$\rho(s) = \rho(s-1) + k_{cp}\{i_{bsse}(s) - i_{bsse}(s-1)\} + k_{ci}i_{bsse} \tag{6}$$

Where, (i_{bsse}) is the error in battery current, which is evaluated from the difference between the reference battery current and the battery observed current [17].

$$i_{bsse}(s) = i^{*}_{bss} - i_{bss}(s) \tag{7}$$

3.3. ICCF-PLL control approach for VSC

An ICCF-PLL control strategy estimates the magnitude of the preliminary load current component. The proposed controls have effective harmonic filtering capability under varying conditions of RESs and load. The ICCF-PLL contains a pair of complex band-pass (CBP) filters, the Centre frequencies of which are at the fundamental negative and positive order frequencies. These band-pass filters work together to estimate the nonlinear load current's fundamental negative and positive sequence components. The obtained FFPS component is passed into the SRF-PLL, which calculates its peak value, frequency, and phase angle. The calculated frequency is given backward towards the complex filters, which are then adjusted to account for proposed system frequency changes.

3.3.1. Modeling of ICCF-PLL

In the proposed ICCF-PLL, the input signals of the $\alpha\beta$-axis are taken as

$$i_{\alpha}(s) = i_1^{+}\cos\left(\theta_1^{+}\right) \tag{8}$$

$$i_{\beta}(s) = i_1^{+}\sin\left(\theta_1^{+}\right) \tag{9}$$

Where, (θ_1^{+}) and (i_1^{+}) represents the phase angle and is its maximum value of the FFPS component of the nonlinear load current. The evaluated components can be described as

$$i^{+}_{\alpha,\,1} = \widehat{i}^{1}_{+}\cos\left(\overrightarrow{\theta}^{+}_{1}\right) \tag{10}$$

$$i^{+}_{\beta,\,1} = \widehat{i}^{1}_{+}\cos\left(\overrightarrow{\theta}^{+}_{1}\right) \tag{11}$$

Where, $(\widehat{i}^{1}_{+})$ and $(\overrightarrow{\theta}^{+}_{1})$ are ideally equal to (i_1^{+}) and (θ_1^{+}), respectively. The phase angle of the obtained FFPS component by the ICCFs can be described using the above assumptions as

$$\widehat{\theta}_1^{+} = \tan^{-1}\left(\widehat{i}^{+}_{\beta,\,1} \Big/ \widehat{i}^{+}_{\alpha,\,1}\right) \tag{12}$$

$$\frac{d\vec{\theta}_1^{+}}{dt} = \frac{\hat{i}_{\alpha,1}^{+}\frac{d\hat{i}_{\beta,1}^{+}}{dt} - \hat{i}_{\beta,1}^{+}\frac{d\hat{i}_{\alpha,1}^{+}}{dt}}{\underbrace{\left(\hat{i}_{\alpha,1}^{+}\right)^2 + \left(\hat{i}_{\beta,1}^{+}\right)^2}_{\left(\hat{i}_{+}^{1}\right)^2}} \tag{13}$$

Where, $\frac{d\hat{i}_{\alpha,1}^{+}}{dt}$ and $\frac{d\hat{i}_{\beta,1}^{+}}{dt}$, according to Fig. 2

$$\frac{d\hat{i}_{\alpha,1}^{+}}{dt} = -\hat{\omega}_g\hat{i}_{\beta,1}^{+} + \omega_p i_\alpha - \omega_p i_{\alpha,1}^{+} - \omega_p\hat{i}_{\alpha,1}^{-} \tag{14}$$

$$\frac{d\hat{i}_{\beta,1}^{+}}{dt} = -\hat{\omega}_g\hat{i}_{\alpha,1}^{+} + \omega_p i_\beta - \omega_p i_{\beta,1}^{+} - \omega_p\hat{i}_{\beta,1}^{-} \tag{15}$$

Substituting (14) and (15) into (13) yields

$$\frac{d\vec{\theta}_1^{+}}{dt} = \hat{\omega}_g + \frac{\omega_p\left(i_{\alpha,1}^{+}i_\beta - i_{\beta,1}^{+}i_\alpha\right)}{\left(\hat{i}_1^{+}\right)^2} \tag{16}$$

It should be noted that the $\alpha\beta$-axis input signals were assumed to be devoid of any FFNS components. As a result, the obtained FFNS components, $(\hat{i}_{\alpha,1}^{-})$ and $(\hat{i}_{\beta,1}^{-})$ are equal to zero in the initial or steady state and have only a minor effect during dynamics.

Substituting (8)–(11) into (16) yields

$$\frac{d\vec{\theta}_1^{+}}{dt} = \hat{\omega}_g + \frac{\omega_p\left(\hat{i}_{\alpha,1}^{+}, i_\beta - \hat{i}_{\beta,1}^{+}, i_\alpha\right)}{\left(\hat{i}_{+}^{1}\right)^2} \tag{17}$$

Assuming that $\theta_1^{+} \approx \vec{\theta}_1^{+}$ and $i_1^{+} \approx \hat{i}_1^{+}$ (17) can be re-written as

$$d\frac{\vec{\theta}_1^{+}}{dt} \approx \hat{\omega}_g + \omega_p\left(\theta_1^{+} - \vec{\theta}_1^{+}\right) \tag{18}$$

The dynamics of the CCF part of the ICCF-PLL are described by Eq. (18), which is in the form equation of linear differential. The entire model of the ICCF-PLL can be achieved using (18) as depicted in Fig. 2.

From this model, the transfer function (open-loop) can be obtained as [30]

$$G_{ol}^{ICCF-PLL} = \frac{\Delta\theta_1(s)}{\Delta\theta_1(s) - \Delta\theta_1(s)} = V_s\frac{\omega_p}{s+\omega_p}\frac{k_p s + k_i}{s^2} \tag{19}$$

Here, v_A represents the maximum value of the terminal voltage, the equation for the same is expressed as

$$v_A = \left(\frac{2}{3}\right)^{1/2}\left(v_{sr}^2 + v_{sy}^2 + v_{sb}^2\right)^{1/2} \tag{20}$$

Here, Eq. (21) shows the process of phase voltages estimation is given as

$$v_{sr} = \frac{1}{3}\left(2v_{sry} + v_{syb}\right), v_{sy} = \frac{1}{3}\left(-v_{sry} + v_{syb}\right), v_{sb} = \frac{1}{3}\left(-v_{sry} + 2v_{syb}\right) \tag{21}$$

Thus, by using Eq. (21), the in-phase and quadrature unit vectors are estimated as

$$u_{rd} = v_{sr}/v_A, u_{yd} = v_{sy}/v_A, u_{bd} = v_{sb}/v_A \tag{22}$$

$$u_{rq} = -\frac{u_{yd}}{\sqrt{3}} + \frac{u_{bd}}{\sqrt{3}}, u_{yq} = -\frac{3u_{bd}}{2\sqrt{3}} + \frac{u_{yd}}{2\sqrt{3}} - \frac{u_{bd}}{2\sqrt{3}}, u_{bq} = -\frac{3u_{rd}}{2\sqrt{3}} + \frac{u_{rd}}{2\sqrt{3}} - \frac{u_{bd}}{2\sqrt{3}} \tag{23}$$

Therefore, the frequency error is found using the difference between the reference frequency (50 Hz) and system frequency provided as

$$f_e(s) = f^*(s) - f(s) \tag{24}$$

The system frequency is evaluated using PLL. Eq. (25) gives the output of the frequency side PI controller

$$i_{fd}(s) = i_{fd}(s-1) + \left\{f_e(s) - k_{pf}f_e(s-1)\right\} + k_{if}f_e(s) \tag{25}$$

Therefore, to determine the total fundamental active current (I_{dT}), the summation of frequency PI controller output and solar PV current is subtracted from the fundamental average load current output (i_{Ldavg}) and the equation is written as

$$I_{dT} = i_{fd} + i_{pv} - i_{Ldavg} \tag{26}$$

Where,

$$i_{Ldavg} = \frac{i_{Ldr} + i_{Ldy} + i_{Ldb}}{3} \tag{27}$$

The fundamental average active component per phase is calculated using all three-phase active components in the above equation. i_{Ldr}, i_{Ldy} and i_{Ldb} are three-phase active load current components in (27). After evaluating the derived load currents and fundamental active components, the sample and hold logic is used. Thus, for phase 'r,' the active load current component (i_{Ldr}), is determined; similarly, for phases 'y' and 'b,' the i_{Ldy} and i_{Ldb} (active load current components) are determined. The reactive component is produced by the voltage PI controller. The equation for this is expressed as

$$i_q(s) = i_q(s-1) + k_{pq}\{v_e(s) - v_e(s-1)\} + k_{iq}\{v_e(s)\} \tag{28}$$

Where,

$$v_e(s) = v_A^*(s) - v_A(s) \tag{29}$$

The amplitude of the supply voltage (v_A) and its reference value at

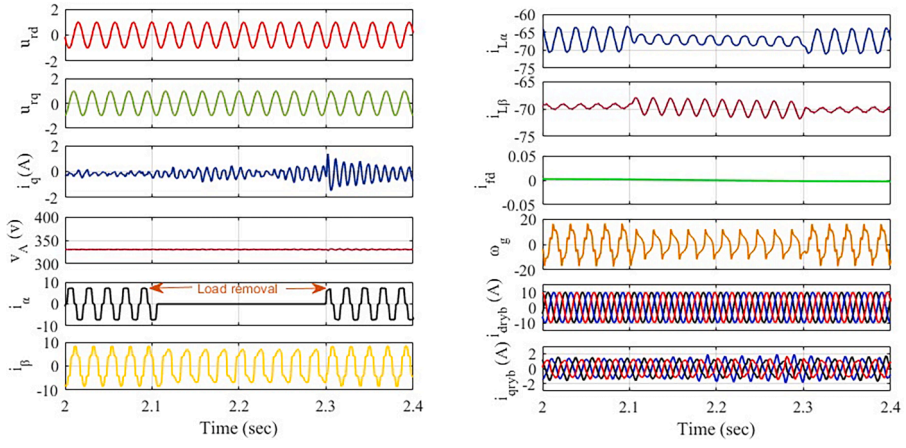


Fig. 3. (a) and (b) Intermediate signals of ICCF-PLL control algorithm.

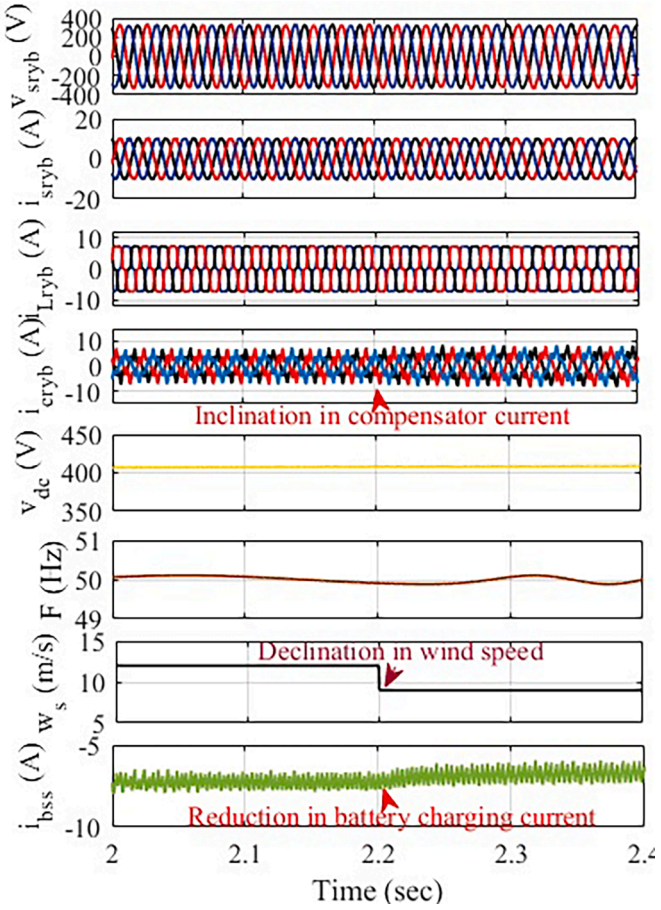


Fig. 4. Response at differing wind speed.

PCI is taken as (v_A^*). So, the total fundamental quadrature component is obtained as

$$I_{qT} = i_q - i_{Lqavg} \tag{30}$$

$$i_{Lqavg} = \frac{i_{Lqr} + i_{Lqy} + i_{Lqb}}{3} \tag{31}$$

From Eqs. (22), (23), (26) and (30) the quadrature and in-phase reference current are computed as

$$i^*_{sdr} = I_{dT} \times u_{rd}, i^*_{sdy} = I_{dT} \times u_{yd}, i^*_{sdb} = I_{dT} \times u_{bd} \tag{32}$$

$$i^*_{sqr} = I_{qT} \times u_{rq}, i^*_{sqy} = I_{qT} \times u_{yq}, i^*_{sqb} = I_{qT} \times u_{bq} \tag{33}$$

The addition of Eqs. (32) and (33) gives the total reference current, which is written as

$$i^*_{sr} = i^*_{sdr} + i^*_{sqr}, i^*_{sy} = i^*_{sdy} + i^*_{sqy}, i^*_{sb} = i^*_{sdb} + i^*_{sqb} \tag{34}$$

The comparison of reference sensed current and observed current generates a current error signal that is further given to the hysteresis controller for the generation of triggering pulses of VSC.

$$i_{sre} = i^*_{sr} - i_{sr}, i_{sye} = i^*_{sy} - i_{sy}, i_{sbe} = i^*_{sb} - i_{sb} \tag{35}$$

4. Results and discussion

The detailed Simulation study and Test results of the DG system comprising of WECS, SPV, and BSS generation for based standalone applications are carried out using MATLAB and an experimental prototype. The isolated system is designed to compensate for the effects of intermittency conditions in RESs and unbalancing in load. Therefore,

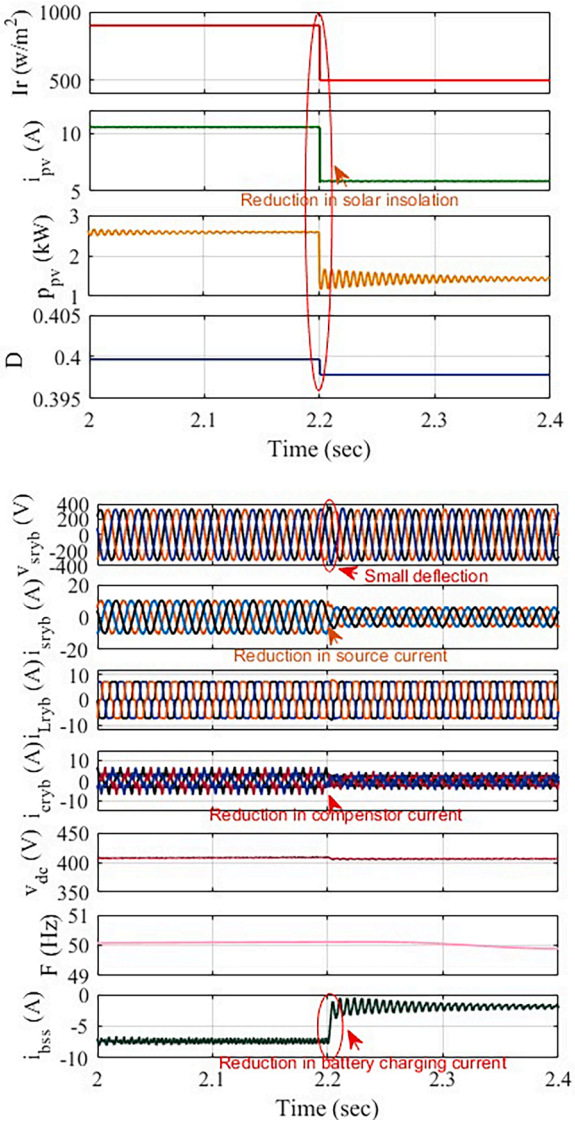


Fig 5. (a) and (b) Response at varying solar insolation.

the BDC control is utilized to control the charging and discharging of BSS for dynamic power balance. Additionally, intermediate control signals of the used ICCF-PLL control algorithms are also presented for a detailed understanding of the DG system operation.

4.1. Intermediate signals of proposed control

Fig. 3(a) and (b) show the intermediate control signals of the ICCF-PLL-based algorithm under unbalanced load conditions. The non-linear load is detached at $t = 2.1$ s to $t = 2.3$ s. Fig. 3(a) represents the signals u_{rd} (active unit templates), u_{rq} (quadrature unit templates), i_q (extracted current from AC PI controller), v_A (voltage amplitude), i_α and i_β (extracted from abc to αβ transformation). The comparison of a reference voltage and sensed voltage yields an AC PI controller to generate the active current component. During the duration of load removal, the extracted i_α and i_β component are reduced to zero. Fig. 3(b) shows the $i_{L\alpha}$, $i_{L\beta}$ (active and reactive load current components), i_{fd} (extracted component from DC PI controller), ω_g (centre frequency), i_{dryb} and i_{qryb} (extracted active and quadrature current components). The comparison of the observed frequency and reference frequency enables the DC PI controller to generate the current component. All the intermediate signals are generated by sensing nonlinear load current (i_{Lryb}).

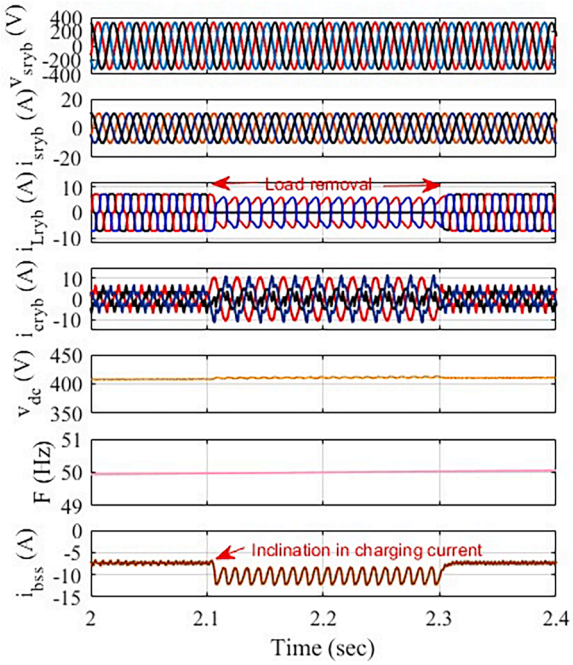


Fig. 6. Dynamic response at unbalanced load.

Further, the intermediate signals are employed to control the standalone DG system.

4.2. Dynamic response of DG system at differing wind speed

Fig. 4 represents the dynamic performance of the DG system under varying wind speeds and fixed solar insolation feeding the nonlinear load. The wind speed differed from 12 m/s to 9 m/s at $t = 2.2$ s. An ICCF-PLL-based controller stabilizes the voltage and DG system frequency at the time of the variation in wind speed. The parameters of a standalone system such as frequency and DC-link voltage are almost constant. The compensator current is marginally rising after the removal of load at $t =$ 2.2 s. The BDC controller manages the process of charging and discharging the BSS and controls the DC-link voltage. During the reduction in wind speed, the BSS is shifted to discharging mode and supplies the current at PCI through VSC to manage the frequency stabilization. Further, an ICCF-PLL control with VSC provides abatement of harmonics, power support and improves the overall PQ of the standalone DG system.

4.3. Dynamic response of DG system under differing solar insolation

Fig 5(a) and (b) represent the dynamic performance of isolated microgrid systems at differing solar insolation and fixed wind speed feeding the constant nonlinear load. In this, the solar insolation is shifted from 900w/m^2 to 500 w/m^2 at $t = 2.2$ s and wind speed is fixed at 12 m/s with constant load. Fig. 5(a) shows due to the reduction in solar insolation, the PV current is reduced from 10.52 A to 5.86 A, the PV power or solar power is declined from 2.5 kW to 1.4 kW and the duty cycle is also reduced. Hence, the total power of the isolated system is reduced to 3.48 kW from 5.83 kW. Fig 5(b) shows due to the sudden change in solar insolation at $t = 2.2$ s, a small deflection in source voltage of 0.022 s is observed after that the proposed controller efficiently controls the source voltage. The source current and the compensator current are reduced according to a change in solar insolation at $t = 2.2$ s. An ICCF-PLL-based controller maintains the system frequency constant during an abrupt change in solar insolation. Therefore, the BDC controller controls the DC-link voltage and shifts the BSS to discharging mode. The stored power in BSS is used to preserve the power balance between the RESs and the nonlinear load. Hence, the controller with the help of VSC contributes voltage and frequency stabilization at PCI, power support, eradication of harmonics, load balancing, and minimizing the PQ problems.

4.4. Dynamic response at an unbalanced load

Fig. 6 represents the dynamic response of the standalone DG system under constant wind speed/solar insolation feeding the unbalanced distorted load. A load of phase 'r' is eliminated at $t = 2.2$ s to $t = 2.4$ s. During the sudden change in load, the load current of the isolated system is reduced and the compensator current is increased. The total generated power of the isolated system is increased to 6.13 kW from 5.83 kW due to the sudden fall of load. During the sudden variation in load at $t = 2.2$ s to $t = 2.4$ s, an ICCF-PLL-based controller successfully maintains the DG system frequency and stabilizes the voltage & current profile at PCI. The BSS charging current is suddenly increased during the disconnection of the load of phase 'r' at $t = 2.2$ s and during the insertion of load at $t = 2.4$ s, the BSS charging current is reduced. Moreover, the BDC controller effectively coordinates the DC-link voltage and manages the charging/discharging modes of BSS. The extra is supplied to BSS during the load rejection process and the stored power of BSS is utilized during the intermittent condition of RESs and load.

4.5. Harmonic spectrum

Fig. 7(a)–(c) represents harmonics spectra and waveform of phase 'r' of the standalone DG system. An ICCF-PLL control-based DG system is used for standalone applications. The THD of the PCI (voltage & current), and load current are found 1.29 %, 4.6 %, and 25.07 % respectively. At steady-state, the PCI currents are sinusoidal with minimum harmonic distortion. The harmonic spectrum of the PCI (voltage & current), and load current of phase 'r' in polluted load condition with a THD of 25.07 % is shown in Fig. 7(a)–(c), Also, the THDs of the PCI currents are well suited within the required range of 5 % due to VSC control.

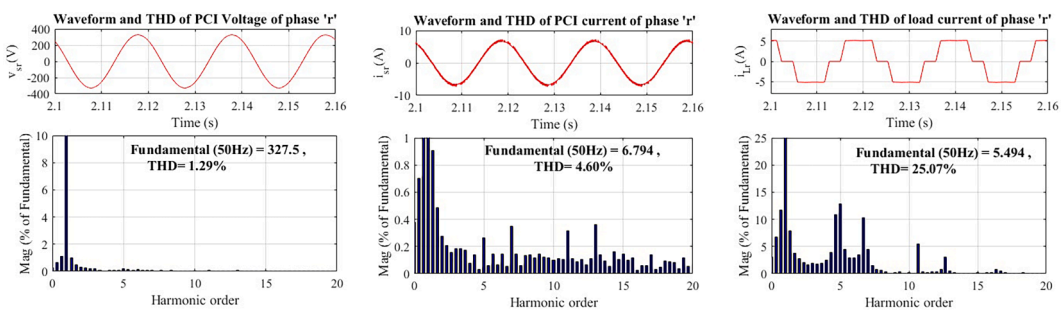


Fig. 7. (a)–(c) Harmonic spectrum.

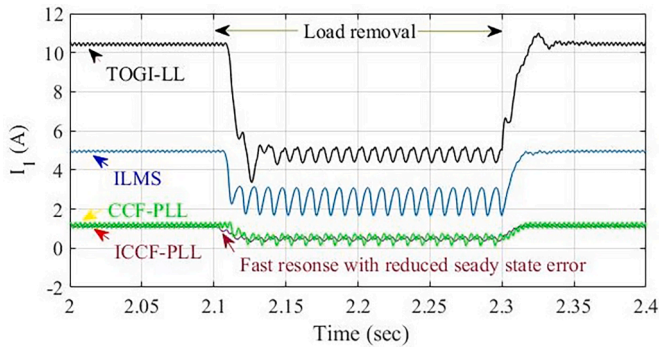


Fig. 8. Comparative assessment of ICCF-PLL control with conventional control algorithms.

Table 1
Comparative performance of control algorithms.

S.no.	Parameters	ICCF-PLL	CCF-PLL	ILMS	TOGI-PLL
1	Settling time	0.02	0.02	0.03	0.04
2	THD obtained	4.17	4.6	4.66	4.8
3	Stability	Best	Better	Good	Good
4	Static error	Very less	less	Large	Medium
5	Oscillations	Very less	less	Large	Medium

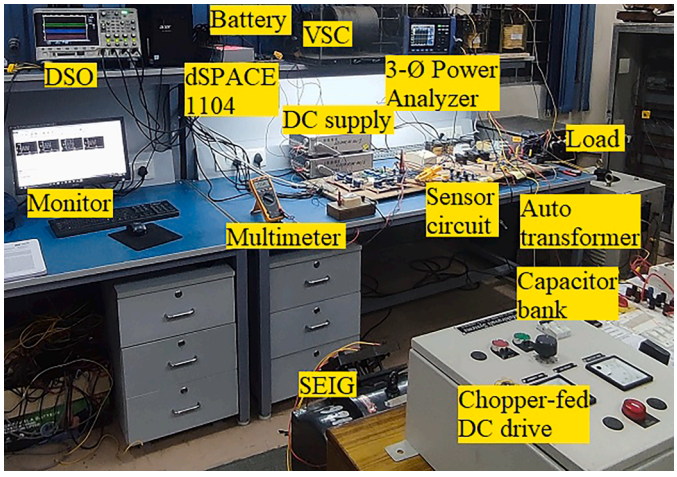


Fig. 9. Control diagram of DG system.

4.6. Comparative assessment of ICCF-PLL control with existing control algorithms

The proposed ICCF-PLL control provides a fast dynamic response with reduced oscillations under the fitful condition of RESs and load dynamics. Fig. 8 depicts the load being rejected of phase 'r' at $t = 2.1$ s and injected at $t = 2.3$ s. When one phase of a partial load is temporally removed and applied, i.e., the extracted fundamental component must first acquire its appropriate value in comparison to the existing control. A frequency-adaptive ICCF-PLL control technique, on the other hand, gives an accurate computation of the fundamental component during dynamic situations. The TOGI-PLL and ILMS have substantial oscillations during load rejection and the dynamic response is excessively slow, while the reaction time is 0.02 s longer in both cases (Load removal and injection). But, the ICCF-PLL response time is quite high and provides a fundamental load current component with less oscillation. Whereas, the ICCF-PLL-based control provides a fast dynamic response and the least steady error with reduced oscillations and extracts the fundamental load current component under different operating conditions.

Table 1 shows the comparative performance of control algorithms. Fig. 8 shows the performance of the ICCF-PLL control approach with CCF-PLL, ILMS, and TOGI-PLL algorithms while extracting the amplitude of the fundamental component. The settling and rise time of the ICCF-PLL approach is less than one cycle after introducing load dynamics. Moreover, the proposed control has fewer oscillations with minimal error, the least harmonics, and the best stability while feeding the dynamic nonlinear load.

5. Test results

A hardware prototype has been formed to examine the system performance in the laboratory under various conditions using the ICCF-PLL-based control algorithm shown in Fig. 9. A VSC, interface and filtering inductors, an optocoupler, connecting cables, and nonlinear loads are included in the proposed hardware setup. The LEM LA-25-based sensors are used to detect PCI current and load side current in standalone DG system phases 'r' and 'y.' The sensor output is forwarded to the DSP-dSPACE 1104's ADC channel through the use of appropriate buffer circuitry.

The dynamic behavior of the DG system was examined using DSO-X 2024A. The harmonics compensating capabilities of the VSC keep the THDs of the PCI currents well under the specified limit. The obtained test results of the DG system have been represented in Figs. 10(a)–(d), 11(a)–(f), and harmonic spectra in 12(a)-(b). The dynamic behavior of the proposed ICCF-PLL approach is given below.

5.1. Intermediate signals of proposed ICCF-PLL control approach

Fig. 10(a)–(d) reveals the intermediate signals of the ICCF-PLL control approach during load dynamics. Fig. 10(a) and (b) reveal load current (i_{Lr}), in-phase unit templates (u_{rd}), quadrature unit templates (u_{rq}) and reference current (i^*_{sr}) of phase 'r' after introducing load dynamics. The load current and reference current reach to zero during the load removal and the unit templates are maintained sinusoidal by the proposed control during the load dynamics. Fig. 10(c) and (d) represent the load current (i_{Lr}), extracted component (i_α and i_β) and voltage amplitude (v_A). The extracted component (i_β) is 90° phase from (i_α). The load current and extracted components are increased or decreased during the dynamic condition of the load and the voltage amplitude is maintained constant during load dynamics by the proposed control approach. The proposed ICCF-PLL control algorithm helps to maintain all internal parameters and regulates the system frequency and voltage of the proposed isolated DG system under all operating conditions.

5.2. Dynamic performance of isolated DG system

The dynamic response of the proposed DG system during dynamic load conditions is depicted in Fig. 11(a)–(f). During load removal, the load current of phase '*r*' and the CPI current of all phases are depicted in Fig. 11(a). The load current and source current is decreased during loading reduction. On the other hand in Fig. 11(b), the load current and source current of all phases is increased during the process of load insertion. The observed test results reveal that the generated current is balanced and sinusoidal even when the load is nonlinear and dynamic. The parameters of the experimental prototype are taken as line voltage (V_{LL}) = 110V, Freq. (f) = 50 Hz, and DC-link voltage = 200 V. During the declination in load, the load current of phase 'r' is reduced to 4.2A from 5.3A, and after load insertion, the load current is increased from 4.2 A to 5.3 A, and the PCI current of all the phases is also reduced to 4.2A from 5.3A during load declination and increased from 4.2 A to 5.3 A after load insertion as shown in Fig. 11(a) and (b). Fig. 11(c) and (d) show the response of the PCI voltage of phase 'r' and the CPI current of all the phases under the transient condition of the load. During the load removal and insertion, the control algorithm stable both CPI voltage &

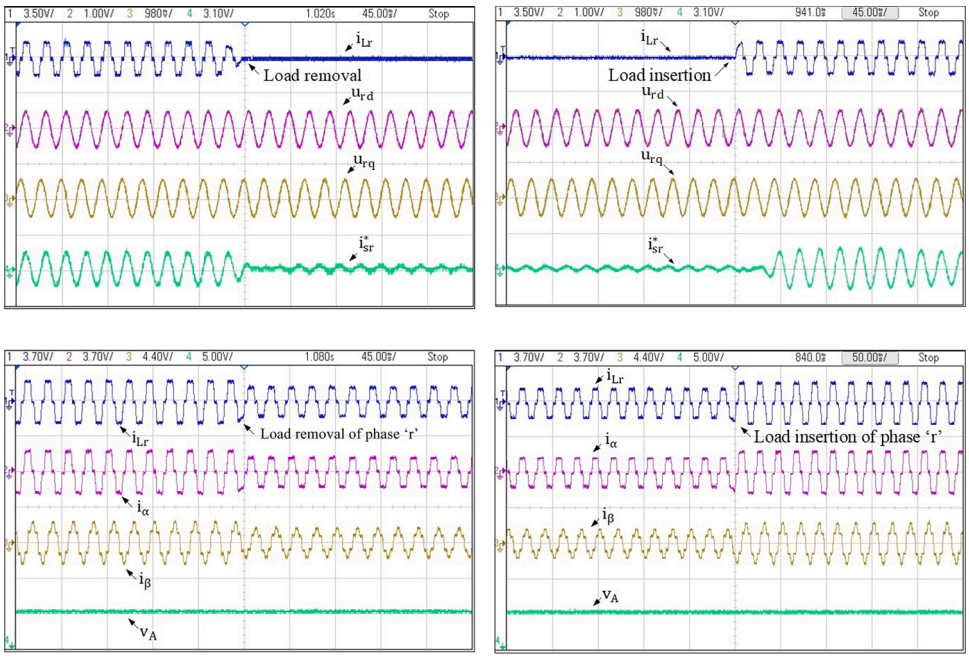


Fig. 10. (a)–(d) Intermediate signals of (a) & (b) i_{Lr}, u_{rd}, u_{rq} and i^{*}_{sr}, and (c) & (d) i_{Lr}, i_{α}, i_{β} and v_A.

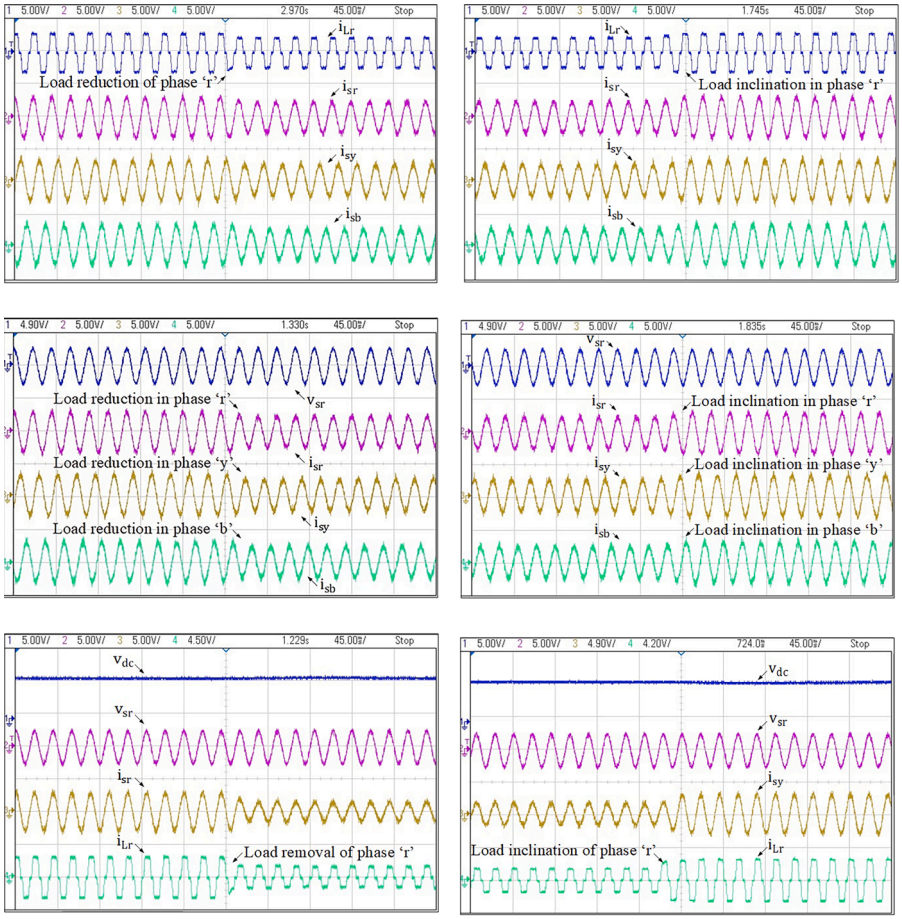


Fig. 11. (a)–(f) Dynamic response at unbalanced load (a)-(b) i_{Lr}, i_{sr}, i_{sy} and i_{sb}, (c)-(d) v_{sr}, i_{sr}, i_{sy} and i_{sb}, (e)-(f) v_{dc}, v_{sr}, i_{sr} and i_{Lr}.

current. Fig. 11(e) and (f) represent the DC-link voltage is settled at 200 V, the PCI current is changed from 5.2 A to 3.2 A during the disconnection of load of two phases and the CPI current is increased from 3.2 A to 5.2 A during the insertion of load. The CPI current and load current are changed with the variation in load. The extra power is supplied to the battery under reduced load demand. The recorded result shows that

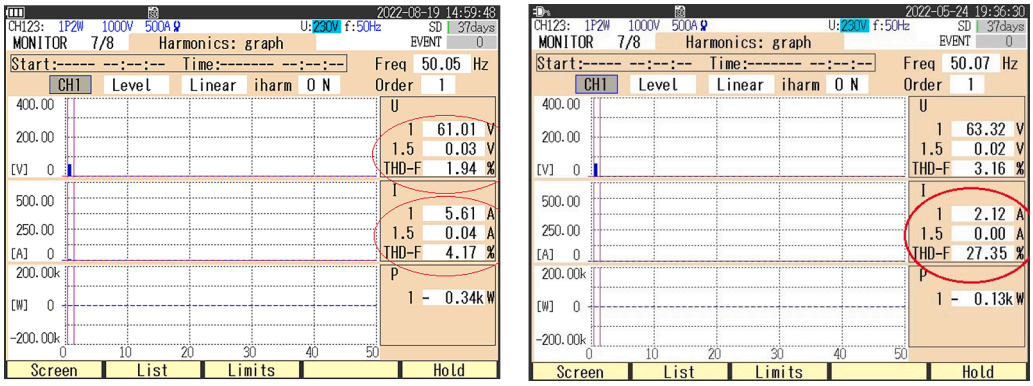


Fig. 12. (a) and (b) Harmonic spectra of DG system (a) PCI current (i_{sr}) and (b) load current (i_{Lr}).

the PCI voltage & current are maintained sinusoidal by the proposed ICCF-PLL control during unbalanced load conditions. Moreover, the proposed control with VSC provides harmonics mitigation, power support, load balancing, and voltage stabilization with reduced overall PQ problems.

5.3. Harmonic spectra of DG system

The harmonics spectrum with waveforms of the proposed standalone DG system is analyzed in detail. The test results validate that THD of the PCI current (i_{sr}), is 4.17 %, and is within the standard limit of the IEEE 519 when connected to a nonlinear load with a current of 27.35 % THD as shown in Fig. 12. The required power support is provided by the battery system through VSC during dynamic conditions of load and wind speed. The proposed control with VSC maintains the power equilibrium and synchronization between the load and RESs.

6. Conclusion

An ICCF-PLL algorithm is used to control the Wind-SolarPV-BSS-based DG system. The proposed standalone DG system is validated using simulation and hardware results to justify its performance under dynamic and steady-state conditions such as unbalanced load, solar intensity change, and wind speed change. The proposed ICCF-PLL approach has estimated the maximum value of fundamental component from load currents to maintain balanced & sinusoidal source currents during dynamic conditions of RESs and load. During transient conditions, this ICCF-PLL control algorithm detects and adapts to frequency variation. Maximum power is extracted and utilized by the SPV, using the MP&O-based MPPT technique. Furthermore, the BSS has effectively controlled the unused power and supply during the dynamic conditions, ensuring stable operation of the islanded DG system across a broad range of operations. Simulation and hardware results show that the ICCF-PLL approach with VSC is effective in load leveling, abatement of harmonics, compensation of reactive power, and PQ improvement. The standalone DG system results, particularly the voltage and source currents at CPI are verified according to the IEEE-519 standard.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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